

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.0      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(zoo)
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##      as.Date, as.Date.numeric
```

```
library(trend)
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
here(); getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
mytheme <- theme_gray() +
  theme(plot.title = element_text(hjust = 0.5))
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1
Garinger_2010 <- read.csv(
  here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv"),
  stringsAsFactors = TRUE)
Garinger_2011 <- read.csv(
  here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv"),
  stringsAsFactors = TRUE)
Garinger_2012 <- read.csv(
  here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv"),
  stringsAsFactors = TRUE)
Garinger_2013 <- read.csv(
  here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv"),
  stringsAsFactors = TRUE)
Garinger_2014 <- read.csv(
  here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv"),
  stringsAsFactors = TRUE)
Garinger_2015 <- read.csv(
  here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv"),
  stringsAsFactors = TRUE)
Garinger_2016 <- read.csv(
  here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv"),
  stringsAsFactors = TRUE)
Garinger_2017 <- read.csv(
  here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv"),
```

```

  stringsAsFactors = TRUE)
Garinger_2018 <- read.csv(
  here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv"),
  stringsAsFactors = TRUE)
Garinger_2019 <- read.csv(
  here("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv"),
  stringsAsFactors = TRUE)

Garinger_complete <- rbind(
  Garinger_2010,
  Garinger_2011,
  Garinger_2012,
  Garinger_2013,
  Garinger_2014,
  Garinger_2015,
  Garinger_2016,
  Garinger_2017,
  Garinger_2018,
  Garinger_2019)

```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```

# 3
Garinger_complete$Date <- mdy(Garinger_complete$Date)

# 4
Garinger_wrangled <- select(Garinger_complete,
  Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)

# 5
Days <- as.data.frame(seq.Date(from = as.Date("2010/01/01"),
  to = as.Date("2019/12/31"),
  by = "day"))

colnames(Days) <- "Date"

# 6
GaringerOzone <- left_join(Days, Garinger_wrangled)

```

```
## Joining with 'by = join_by(Date)'
```

```
colnames(GaringerOzone) <- c("Date", "O3ppm", "AQI")
summary(GaringerOzone) #checking for NAs
```

```
##           Date           O3ppm           AQI
## Min.      :2010-01-01   Min.      :0.00200   Min.      : 2.00
## 1st Qu.:2012-07-01   1st Qu.:0.03200   1st Qu.: 30.00
## Median :2014-12-31   Median :0.04100   Median : 38.00
## Mean      :2014-12-31   Mean      :0.04163   Mean      : 41.57
## 3rd Qu.:2017-07-01   3rd Qu.:0.05100   3rd Qu.: 47.00
## Max.      :2019-12-31   Max.      :0.09300   Max.      :169.00
##                                     NA's      :63
```

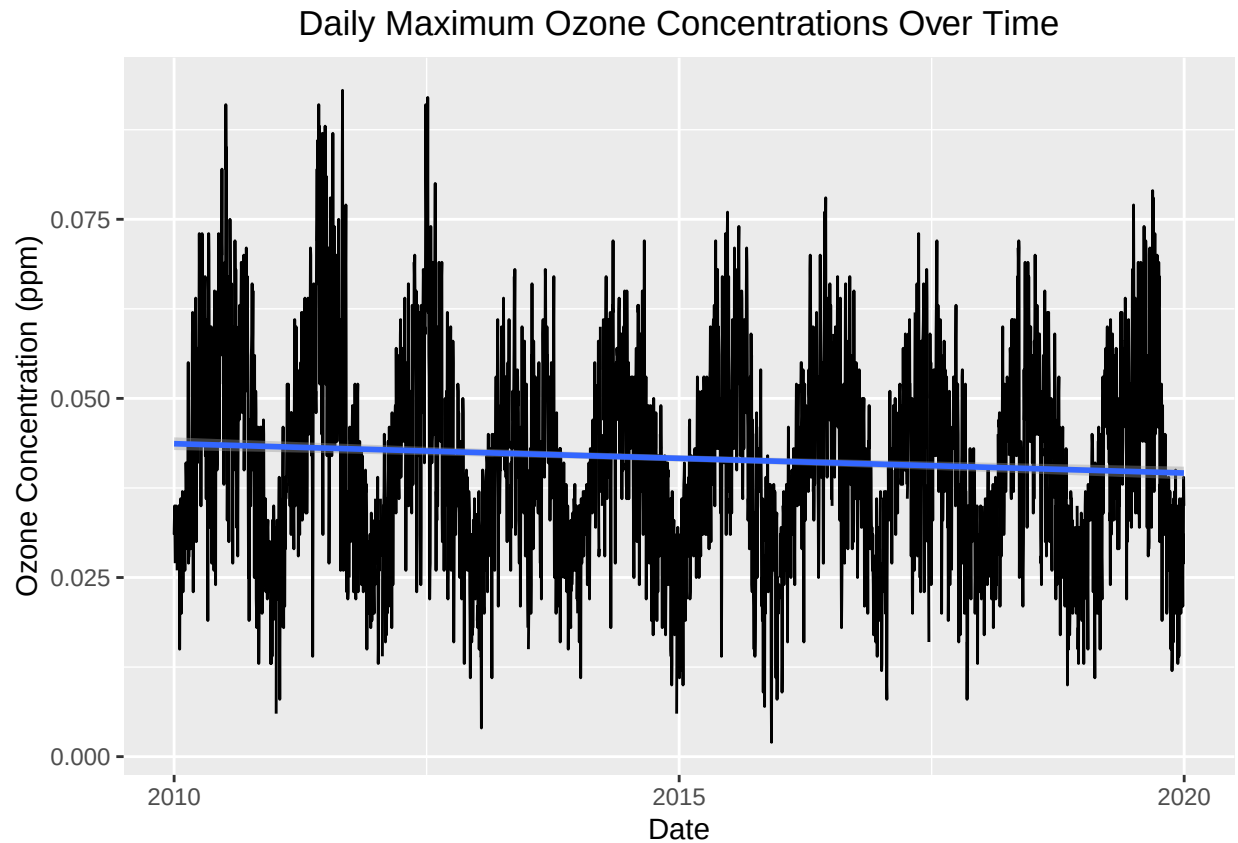
Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7
ggplot(GaringerOzone,
       aes(x = Date, y = O3ppm)) +
  geom_line() +
  geom_smooth(method = "lm") +
  labs(
    y = "Ozone Concentration (ppm)",
    title = "Daily Maximum Ozone Concentrations Over Time")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: The plot suggests a slight downward trend over time.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
GaringerOzone.clean <-
  GaringerOzone %>%
    mutate(O3ppm = na.approx(O3ppm))
summary(GaringerOzone.clean) #checking for NAs
```

```
##      Date              O3ppm              AQI
##  Min.   :2010-01-01   Min.   :0.00200   Min.   :  2.00
## 1st Qu.:2012-07-01   1st Qu.:0.03200   1st Qu.: 30.00
##  Median:2014-12-31   Median :0.04100   Median : 38.00
##   Mean  :2014-12-31   Mean   :0.04151   Mean   : 41.57
## 3rd Qu.:2017-07-01   3rd Qu.:0.05100   3rd Qu.: 47.00
##   Max.  :2019-12-31   Max.   :0.09300   Max.   :169.00
##                                     NA's   :63
```

Answer: Linear interpolation is best here because the data has seasonality and data gaps are relatively small. Using linear connections maintains the direction of the seasonal component and produces insignificant error. Piecewise interpolation would stagnate sections of the data, accentuating error in the context of the seasonal component. Spline interpolations uses a quadratic function to produce smooth curves; this would not resemble the behavior of the rest of the data.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9
GaringerOzone.monthly <-
  GaringerOzone.clean %>%
    mutate(Month = month(Date), Year = year(Date)) %>%
    group_by(Year, Month) %>%
    summarise(MeanO3ppm = mean(O3ppm)) %>%
    mutate(Date = my(paste0(Month, "-", Year)))
```

```
## 'summarise()' has grouped output by 'Year'. You can override using the
## '.groups' argument.
```

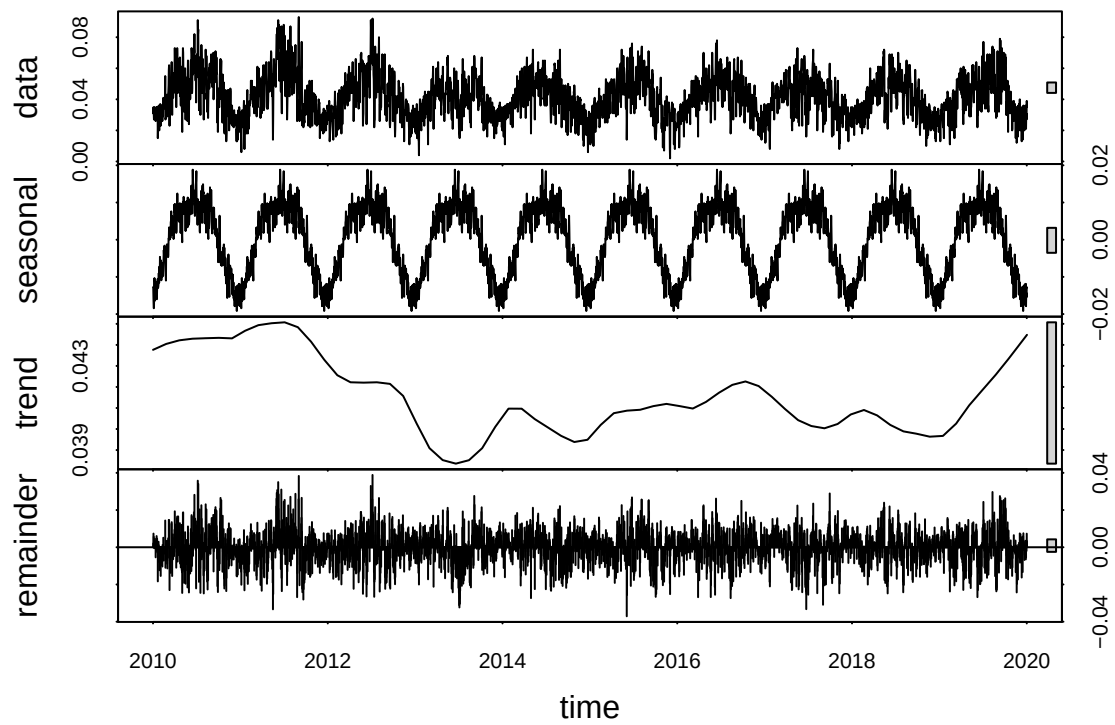
10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

```
#10
start_date <- min(GaringerOzone.clean$Date)
GaringerOzone.daily.ts <- ts(
  GaringerOzone.clean$O3ppm,
  start = c(year(start_date), month(start_date), day(start_date)),
  frequency = 365)

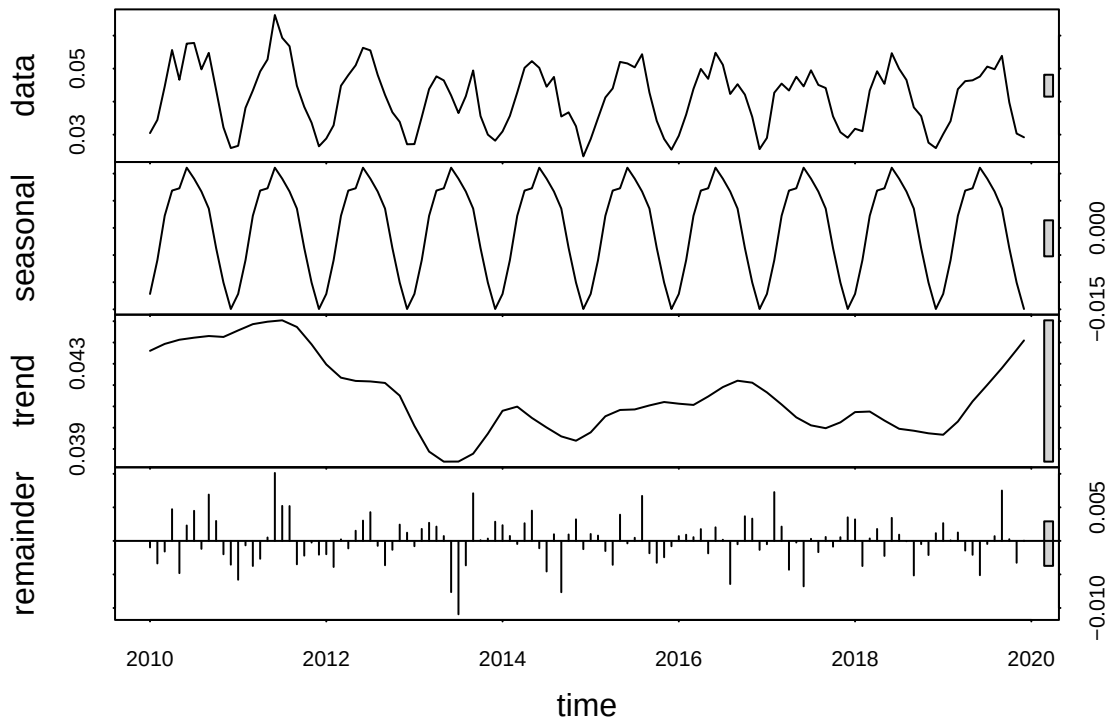
start_month <- min(GaringerOzone.monthly$Date)
GaringerOzone.monthly.ts <- ts(
  GaringerOzone.monthly$MeanO3ppm,
  start = c(year(start_month), month(start_month)),
  frequency = 12)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11
GaringerOzone.daily.decomposed <- stl(
  GaringerOzone.daily.ts,
  s.window = "periodic")
plot(GaringerOzone.daily.decomposed)
```



```
GaringerOzone.monthly.decomposed <- stl(
  GaringerOzone.monthly.ts,
  s.window = "periodic")
plot(GaringerOzone.monthly.decomposed)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

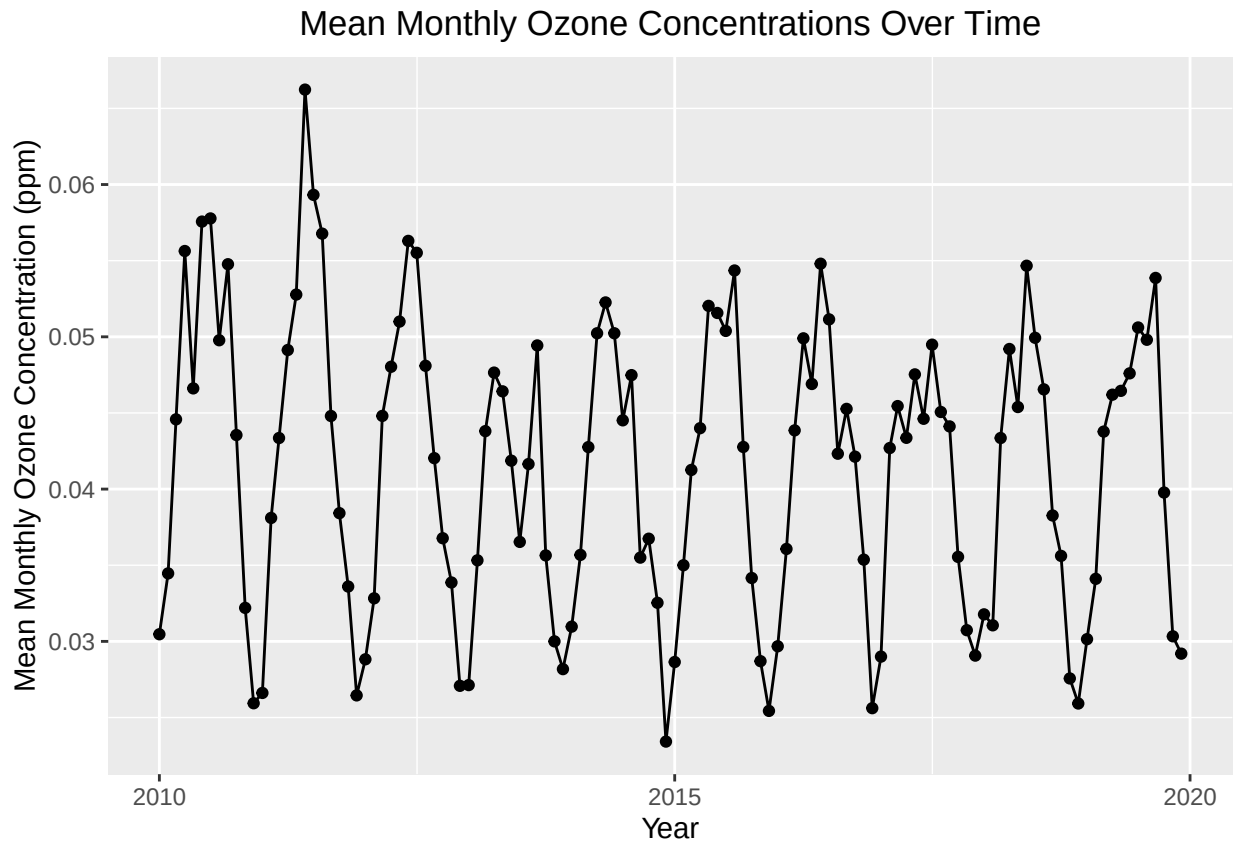
```
#12
smk.test(GaringerOzone.monthly.ts)

##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## z = -1.963, p-value = 0.04965
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##      S varS
## -77 1499
```

Answer: The decomposed time series plots indicate a strong seasonal component in these data. Only the seasonal Mann-Kendall analysis can be run on data with seasonality. Other methods, like linear regression, Mann-Kendall, and Spearman Rho, will not produce accurate results if seasonality is not removed.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.


```
# 13
ggplot(GaringerOzone.monthly,
       aes(x = Date, y = MeanO3ppm)) +
  geom_point() +
  geom_line() +
  labs(
    x = "Year",
    y = "Mean Monthly Ozone Concentration (ppm)",
    title = "Mean Monthly Ozone Concentrations Over Time")
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: The ozone concentrations from 2010 to 2020 show strong seasonality, which appears clearly in the monthly mean plot as periodic fluctuations. The Seasonal Mann-Kendall test reveals a statistically significant downward monotonic trend in mean monthly ozone concentrations over this period ($S < 0$, $p\text{-value} = 0.04965$).

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```

#15
Ozone.Monthly.seasonal.ts <-
  GaringerOzone.monthly.decomposed$time.series[, "seasonal"]

Ozone.Monthly.deseason.ts <-
  GaringerOzone.monthly.ts - Ozone.Monthly.seasonal.ts

#16
mk.test(Ozone.Monthly.deseason.ts)

```

```

##
## Mann-Kendall trend test
##
## data: Ozone.Monthly.deseason.ts
## z = -2.672, n = 120, p-value = 0.00754
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##          S          varS          tau
## -1.179000e+03  1.943657e+05 -1.651376e-01

```

Answer: Both tests produce a p-value below 0.05, agreeing that the null hypothesis of stationarity can be rejected. In other words, the mean monthly ozone concentrations have a monotonic trend over time. The trend is downward according to the negative S values of both tests. The Seasonal Mann-Kendall test shows slight statistical significance (p-value = 0.04965), while the Mann-Kendall test on non-seasonal data shows stronger statistical significance (p-value = 0.00754).